Mathematical Modeling Lecture Notes
Chapter 1 – Modeling Change
(Discrete Dynamical Systems)

I. Introduction
   a. Background
      i. Many situations that you might want to model have a time component, and aspects of the situation change over time.
      ii. One way to get a handle on this is to focus on the change.
      iii. For example, a simple model for handling situations with a time component is the following: *future value = present value + change.*
      iv. Solving for the change yields *change = future value − present value.*
      v. If a_n represents the value of some quantity at time n, then this equation could be represented as $\Delta a_n = a_{n+1} - a_n$.
      vi. And if we could find an expression for Δa_n, or experiment with different expressions for Δa_n, then we could see how the value of the quantity changes over time.
   b. Difference equations
      i. An equation of the form $f(n, a_0, a_1, \ldots, a_n) = a_{n+1}$ is called a *difference equation.*
      ii. For example, $1 = a_{n+1} - a_n$ is a difference equation.
      iii. Difference equations are similar to differential equations.
         1. If you view the input variable as a discrete quantity, then you end up with a difference equation.
         2. If you view the input variable as a continuous quantity, then you end up with a differential equation.
         3. The techniques for solving difference equations and differential equations have some similarities.
      iv. An advantage of difference equations is it's easy to apply simulation to them.
         1. If you have an expression for Δa_n and you know what a_0 and any other initial conditions are, then you can program a computer to calculate as many values of a_n as you want quite quickly.
         2. So you can watch the values of a_n change over time.
      v. You can also have coupled difference equations, just like you have coupled differential equations.
         1. For example, perhaps you have two systems that you want to keep track of, $\{a_n\}$ and $\{b_n\}$.
         2. Perhaps Δa_n and Δb_n are functions of both a_n and b_n.
         3. A real-world example of this is a predator-prey system.
   c. Looking at a quantity from a discrete dynamical system point of view is a classical computer scientist's perspective.
      i. Nearly everything computer scientists deal with are discrete.
      ii. Computers model events changing discretely over time quite easily.

II. A simple example: Compound interest
   a. A bank account with interest compounded periodically is a classic example of a simple discrete dynamical system.
   b. Suppose the account earns 3% interest per year, compounded annually.

c. Deriving the difference equation
   i. Then Δa_n, the amount of interest added to the account at the end of the year, is $0.03a_n$, 3% of whatever was in the account at the beginning of the year.
   ii. This gives us the equation $0.03a_n = a_{n+1} - a_n$, or $a_{n+1} = 1.03a_n$.
   iii. In general, we would have $a_{n+1} = (1 + r)a_n$, where r is the interest rate.

d. Show data and graph to students in Python. (File in Users/mspivey/Math471/.)

e. Solving the difference equation
   i. This equation is simple enough that we can solve it directly; i.e., obtain a_n in terms of a_0 and n.
   ii. We have $a_1 = (1 + r)a_0$.
   iii. And $a_2 = (1 + r)a_1 = (1 + r)^2 a_0$.
   iv. An induction proof yields $a_n = (1 + r)^n a_0$.

III. A more complicated example: Fishing

a. Overfishing is a real problem in certain areas of the world, such as the North Sea in Europe, the Grand Banks off the eastern coast of Canada, and (I believe) our own Puget Sound.

b. What would be the major changes to the number of fish in a particular region in a given year?
   i. Natural population growth
   ii. Fish harvesting

c. Initial model
   i. The change in the fish population in year n could then be $\Delta a_n = ka_n - d$, where k is a growth factor related to the number of fish and d is the number of fish harvested per year.
   ii. This gives us the difference equation $a_{n+1} = (1 + k)a_n - d$.
   iii. For example, suppose $a_0 = 1000$, $k = 0.3$, and $d = 100$.
   iv. Play around with different values of a_0, k, and d with Python.
      1. Show different situations in which the fish population dies out, stabilizes, or grows without bound.
      2. For example, changing to $d = 300$ and keeping the other parameters the same results in an equilibrium value.

d. A second attempt
   i. It's a bit simplistic to assume that the fish population might grow without bound.
   ii. We can make our model more realistic by including a factor that limits the growth of the population.
   iii. The standard way to do that is to modify that growth factor so that it's proportional both to the current number of fish in the population and how far the current population is away from the maximum population.
   iv. This leaves us the equation $\Delta a_n = ka_n(1-a_n/M)-d$, where M is the maximum population.
   v. And the difference equation becomes $a_{n+1} = a_n + ka_n(1 - a_n/M) - d$.
   vi. This is an example of a nonlinear difference equation.
   vii. Let $M = 2500$ initially.
   viii. Play around with different values of a_0, k, d, and M in Python.
      1. Once again, show different situations in which the fish population dies out or stabilizes.

2. In particular, note how changing any of the parameters (initial population, growth rate, number of fish harvested, and even maximum population) can have a drastic effect on the final fish population.
3. For the initial parameters of the other values, $d = 180$ is an equilibrium solution.

e. Other fish models? (Ask students.)

IV. Additional problems to try to model (Give to students.)

a. Consider the spreading of a highly communicable disease on an isolated island with population size N. A portion of the population travels abroad and returns to the island infected with the disease. Formulate a dynamical system to approximate the change in the number of people in the population who have the disease.
   i. The best model is probably $\Delta a_n = k\frac{a_n}{N}\left(1 - \frac{a_n}{N}\right)N = ka_n(1 - a_n/N)$, although other options are possible.
   ii. The *law of mass action* states that the rate at which two populations interact is proportional to the product of their sizes, and we see that with this model.

b. The rhinoceros is an endangered species. Suppose enough game preserve land is set aside so that there is sufficient room for many more rhinoceros territories than there are rhinoceroses. Consequently, there will be no danger of overcrowding. However, if the population is too small, fertile adults will have difficulty finding each other when it comes time to mate. Write a difference equation that models the rhinoceros population.
   i. One possibility: $\Delta a_n = ka_n(a_n/M - 1)$, where M is the minimum required threshold.
   ii. Another possibility: $\Delta a_n = ka_n - b$, where b is a parameter that determines the level at which the population starts to decrease.

V. Checking a model

a. We've discussed some models; how good are they really?
b. Let's take a set of data and fit a model to it.
c. Stopping distance problem (1.2 #9)
   i. How do speed of an automobile and stopping distance relate? Ask students.
   ii. The data below show the speed n in increments of 5 mph of an automobile and the associated distance a_n in feet required to stop it once the brakes are applied. For instance, $n = 6$ (30 mph) requires a stopping distance of $a_6 = 47$ feet.

n	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
a_n	3	6	11	21	32	47	65	87	112	140	171	204	241	282	325	376

   iii. Calculate and plot the change Δa_n versus n. (Do with Python; use *diff* command.)
   iv. Does the graph reasonably approximate a linear relationship? Yes.
   v. Now, find a difference equation model for the stopping distance data. Test your model by plotting the errors in the predicted values against n.
      - A fairly decent model is $\Delta a_n = 3.1n$.
      - Averaging the second finite differences gives $\Delta a_n = 3.4n$ (approximately).
   vi. Discuss the appropriateness of the model.

VI. Long-term behavior

a. Generally what we care about with a discrete dynamical system is the long-term behavior of the system.
b. Of particular interest are the *equilibrium values* or *fixed points* of the system.
    i. These are the values with the property that if the dynamical system ever reaches that value, then it stays at that value for all time.
    ii. An equilibrium value has the property that if a_n is ever equal to the equilibrium value then $a_{n+1} = a_n$.
    iii. So we can solve for the equilibrium value(s) by replacing a_{n+1} by a_n.
    iv. Equivalently, we can set $\Delta a_n = 0$ and solve.
    v. Types of equilibrium values
        1. If values near an equilibrium value tend to head toward that value over time then the value is said to be *stable*.
        2. If values near an equilibrium value tend not to head toward that value over time then the value is said to be *unstable*.
c. Equations of the form $a_{n+1} = ra_n$
    i. The solution to this equation is $a_n = r^n a_0$.
        1. We can see this from generalizing the compound interest example.
        2. We can prove it by induction.
    ii. Determining the equilibrium points
        1. Substitution yields $a_n = ra_n$.
        2. Thus we have $a_n(1 - r) = 0$.
        3. So $a_n = 0$ or $r = 1$.
        4. Thus $a_n = 0$ is an equilibrium point regardless of the value of r.
        5. And if $r = 1$ then every real number is an equilibrium point.
    iii. What happens in the long run if $r \neq 1$? (Play with this on Python.)
        1. If $r > 1$, then the system grows large (if $a_0 > 0$) or negative (if $a_0 < 0$) without bound.
        2. If $0 < r < 1$, then the system decays monotonically to 0.
        3. If $-1 < r < 0$, then the system oscillates toward 0.
        4. If $r = -1$, then the system oscillates between a_0 and $-a_0$ forever.
        5. If $r < -1$, then then system oscillates forever, growing ever further from 0 in both positive and negative directions.
    iv. The stability of the equilibrium value $a_0 = 0$ depends on the value of r.
d. Equations of the form $a_{n+1} = ra_n + b$
    i. Determining the equilibrium points
        1. Substitution yields $a_n = ra_n + b$.
        2. Thus we have $a_n(1 - r) = b$.
        3. And then $a_n = b/(1 - r)$, provided $r \neq 1$.
        4. If $r = 1$ then there is no equilibrium value.
        5. Initial fish example
            - $r = 1.3$ and $b = -100$
            - Thus there is an equilibrium at $333\,1/3$.
    ii. Long-run behavior
        1. This is a little harder to analyze because of the additional parameter b.
        2. Try examples with Python.
        3. It turns out that
            - If $|r| < 1$ then then $a = b/(1 - r)$ is stable.

- If $|r| > 1$ or $r = -1$ then $a = b/(1-r)$ is unstable.
- If $r = 1$ then there are no equilibrium points.

4. This can be proved with something called the *Banach fixed-point theorem*, which is beyond the scope of this class.

VII. Solution techniques for certain difference equations

a. Introduction

i. Just as with differential equations, certain classes of difference equations admit to relatively easy solution procedures.

ii. We'll concentrate on first- and second-order equations.

b. First-order equations

i. A *first-order linear difference equation* is one of the form $a_{n+1} = p_n a_n + q_n$, where p and q are both functions of n.

ii. Homogeneous equations

1. The first-order linear difference equation $a_{n+1} = p_n a_n + q_n$ is said to be *homogeneous* if $q_n = 0$.
   - For example, the compound interest equation is homogeneous with $p_n = 1.03$.
2. The solution to a homogeneous first-order difference equation is $a_n = C\prod_{i=0}^{n-1} p_i$, with $C = a_0$.
   - We have $a_1 = p_0 a_0 = Cp_0$.
   - And $a_2 = p_1 a_1 = p_1(Cp_0) = Cp_1 p_0$.
   - Continuing in this vein, we have $a_n = C\prod_{i=0}^{n-1} p_i$.
3. For example, suppose we have $a_n = na_{n-1}$, with $a_0 = 1$. Then the solution would be $a_n = n!$.

iii. Nonhomogeneous equations

1. The solution to the nonhomogeneous difference equation $a_{n+1} - p_n a_n = q_n$ is

$$a_n = a_0 \prod_{i=0}^{n-1} p_i + \left(\prod_{i=0}^{n-1} p_i\right)\left(\sum_{i=0}^{n-1} \frac{q_i}{\prod_{j=0}^{i} p_j}\right),$$

provided the expression in the denominator is nonzero.

- Dividing both sides of the difference equation by $\prod_{i=0}^{n} p_i$ yields

$$\frac{a_{n+1}}{\prod_{i=0}^{n} p_i} - \frac{a_n}{\prod_{i=0}^{n-1} p_i} = \frac{q_n}{\prod_{i=0}^{n} p_i}.$$

- This can be expressed as

$$\Delta\left(\frac{a_n}{\prod_{i=0}^{n-1} p_i}\right) = \frac{q_n}{\prod_{i=0}^{n} p_i}.$$

  - The form of this expression is $\Delta b_n = r_n$, or $b_{n+1} = b_n + r_n$.
  - Here we have $b_n = \dfrac{a_n}{\prod_{i=0}^{n-1} p_i}$ and $r_n = \dfrac{q_n}{\prod_{i=0}^{n} p_i}$.
  - With each additional term in the sequence you're adding r_n.
  - So the solution is $b_n = b_0 + \sum_{i=0}^{n-1} r_i$.

- Rewriting, we have

$$\frac{a_n}{\prod_{i=0}^{n-1} p_i} = \frac{a_0}{\prod_{i=0}^{-1} p_i} + \sum_{i=0}^{n-1} \frac{q_i}{\prod_{j=0}^{i} p_j}$$

$$\Longrightarrow a_n = a_0 \prod_{i=0}^{n-1} p_i + \left(\prod_{i=0}^{n-1} p_i \right) \left(\sum_{i=0}^{n-1} \frac{q_i}{\prod_{j=0}^{i} p_j} \right).$$

- Special case $a_{n+1} = ra_n + b$, $r \neq 1$
  - Here, the solution is

$$a_n = a_0 r^n + r^n \sum_{i=0}^{n-1} \frac{b}{r^{i+1}} = a_0 r^n + b r^{n-1} \frac{1 - r^{-n}}{1 - r^{-1}} = a_0 r^n + b r^n \frac{1 - r^{-n}}{r - 1} = a_0 r^n + b \frac{1 - r^n}{1 - r}$$

$$= \left(a_0 - \frac{b}{1 - r} \right) r^n + \frac{b}{1 - r}.$$

  - In this derivation we use the formula for the partial sum of a geometric series: $\sum_{i=0}^{n} r^i = \frac{1 - r^{n+1}}{1 - r}$.
  - Your text has a variant of this solution. (See Theorem 3 on page 30.)

2. Examples
   - Solve the difference equation $a_{n+1} = (n + 1)a_n + 1$.
     - We have $p_n = n + 1$ and $q_n = 1$.
     - Plugging into the formula yields

$$a_n = a_0 n! + \sum_{i=0}^{n-1} \frac{n!}{(i+1)!} = a_0 n! + \sum_{i=1}^{n} \frac{n!}{i!}.$$

   - The first difference equation we considered for the fishing model was the equation $a_{n+1} - a_n = ka_n - d$.
     - This can be written as $a_{n+1} = (1+k)a_n - d$, which fits the special case with $r = 1 + k$ and $b = -d$.
     - Thus the solution is

$$a_n = \left(a_0 + \frac{d}{-k} \right) (1 + k)^n + \frac{d}{k} = a_0 (1 + k)^n + \frac{d}{k} (1 - (1 + k)^n).$$

c. Second-order equations

i. Introduction

1. A second-order difference equation is one in which Δa_n is a function of both a_n and a_{n-1} (and possibly n).
2. The only second-order difference equations we will be considering are those that are linear and homogeneous and have constant coefficients, i.e., those of the form $a_{n+1} = Aa_n + Ba_{n-1}$, where A and B are constants.

ii. Solution approach

1. Basic idea
   - Given what we know about the solutions to first-order equations it makes sense that a solution to a second-order equation with constant coefficients might have the form $a_n = t^n$ for some t.
   - If we substitute that into the homogeneous equation we get $t^{n+1} = At^n + Bt^{n-1}$.

- Dividing by t^{n-1} we obtain $t^2 = At + B$, or $t^2 - At - B = 0$.
- So any solution of the form $a_n = t^n$ must involve an t that satisfies this equation.
- This is generally true; the solution to a second-order equation turns out to depend on the roots of this *characteristic equation* $t^2 - At - B = 0$.

2. Two distinct roots case
   - Theorem: Suppose that the characteristic equation $t^2 - At - B = 0$ for the difference equation $a_{n+1} = Aa_n + Ba_{n-1}$ has two distinct roots r and s. Then $a_n = Cr^n + Ds^n$ is a solution to the difference equation, where C and D are constants such that $a_0 = C + D$ and $a_1 = Cr + Ds$.
   - Proof
     - Since r and s are solutions to the characteristic equation we know that $Ar^{n-1} + Br^{n-2} = r^n$ and $As^{n-1} + Bs^{n-2} = s^n$ for $n \geq 2$.
     - The theorem is clearly true in the cases $n = 0$ and $n = 1$.
     - Fix $k \geq 2$, and suppose that the theorem is true for all values of n less than k.
     - Then

$$\begin{aligned} a_k &= Aa_{k-1} + Ba_{k-2} \\ &= A(Cr^{k-1} + Ds^{k-1}) + B(Cr^{k-2} + Ds^{k-2}) \\ &= C(Ar^{k-1} + Br^{k-2}) + D(As^{k-1} + Bs^{k-2}) \\ &= Cr^k + Ds^k. \end{aligned}$$

     - By induction, then, the result is proved.
   - Example: Fibonacci sequence
     - The Fibonacci sequence is the sequence $0, 1, 1, 2, 3, 5, 8, 13, \ldots$, given by $F_0 = 0$, $F_1 = 1$, and $F_{n+1} = F_n + F_{n-1}$ for all $n \geq 1$.
     - Its characteristic equation is $t^2 - t - 1 = 0$.
     - The solutions are $t = (1 \pm \sqrt{1+4})/2 = (1 \pm \sqrt{5})/2$.
     - The constants C and D must satisfy $C + D = 0$, $C(1+\sqrt{5})/2 + D(1-\sqrt{5})/2 = 1$.
     - These solutions are $C = 1/\sqrt{5}$, $D = -1/\sqrt{5}$.
     - Thus we have

$$F_n = \frac{1}{\sqrt{5}}\left(\frac{1+\sqrt{5}}{2}\right)^n - \frac{1}{\sqrt{5}}\left(\frac{1-\sqrt{5}}{2}\right)^n.$$

     - Believe it or not, all of these values are actually integers!
     - Show students small values in Python.
     - Side note: $(1 + \sqrt{5})/2$ is the golden ratio.

3. One (double) root case
   - The previous result requires that there actually be two distinct roots r and s to the characteristic equation $t^2 - At - B = 0$.
     - What if there is only one root, r?
     - For example, the equation $t^2 - 4t + 4 = 0$ only has one root, 2.
   - However, the sequence $(s_n) = (nr^n) = 0, r, 2r^2, 3r^3, \ldots$ also satisfies the difference equation in this case.

- First, if $t^2 - At - B = 0$ has only the root r, then $t^2 - At - B = (t - r)^2 = t^2 - 2rt + r^2$, which means $A = 2r$ and $b = -r^2$.
- Substituting, we have

$$As_n + Bs_{n-1} = Anr^n + B(n-1)r^{n-1} = 2rnr^n - r^2(n-1)r^{n-1}$$
$$= 2nr^{n+1} - (n-1)r^{n+1} = (n+1)r^{n+1} = s_{n+1}.$$

- Theorem: Suppose that the characteristic equation $t^2 - At - B = 0$ for the difference equation $a_{n+1} = Aa_n + Ba_{n-1}$ has the single root r. Then $a_n = Cr^n + Dnr^n$ is a solution to the difference equation, where C and D are constants such that $a_0 = C$ and $a_1 = (C + D)r$.
  - The proof is similar to that for the two distinct roots case.
- Example: $a_{n+1} = 4a_n - 4a_{n-1}$, with $a_0 = 1$ and $a_1 = 3$.
  - The characteristic equation is $t^2 - 4t + 4 = 0$, which has the single root $t = 2$.
  - Thus the solution is $a_n = C2^n + Dn2^n$, where C and D need to be determined.
  - Clearly, $C = 1$.
  - To find D, solve: $3 = (1 + D)2 \implies D = 1/2$.
  - So the solution is $a_n = 2^n + (1/2)n2^n = 2^n + n2^{n-1}$.

d. For more on solutions of difference equations, see any difference equations text.

VIII. Systems of difference equations

a. Sometimes two quantities that change over time are related; these can give rise to systems of difference equations.

b. Example: Find a system of difference equations that describes the dynamics of a fox population and a rabbit population in a given area.

i. We'll use these assumptions:
   1. If no foxes are present the rabbits grow unchecked at a rate proportional to their current population.
   2. Foxes eat rabbits at a rate proportional to the rate at which the foxes and rabbits interact.
   3. If there are no rabbits the foxes decline at a rate proportional to their population.
   4. The fox growth rate is proportional to the number of rabbits eaten by foxes (which is proportional to the rate at which the foxes and rabbits interact).

ii. The changes can be expressed like so:
   1. $\Delta r_n = Ar_n - Br_nf_n$.
   2. $\Delta f_n = -Cf_n + Dr_nf_n$.

iii. In terms of r_{n+1} and f_{n+1}
   1. $r_{n+1} = (1 + A)r_n - Br_nf_n$.
   2. $f_{n+1} = (1 - C)f_n + Dr_nf_n$.

iv. Graph these for various values of A, B, C, and D and initial populations.
   - Start with $A = C = 0.2$, $B = 0.001$, $D = 0.002$, and initial populations of 100 each.
   - Also do plot of foxes vs. rabbits.

v. Find the equilibrium points for the system with the previous parameters.
   1. We have $0 = 0.2r - 0.001rf$ and $0 = -0.2f + 0.002rf$.

2. The first equation yields $r = 0$ or $f = 200$.
3. The second equation yields $f = 0$ or $r = 100$.
4. This yields the equilibrium solutions (0,0) and (100,200).
5. Graph systems with equilibrium values as the initial values.
6. Do perturbations of these (5-unit, 10-unit, 20-unit).

c. Example: The leading brewery on the West Coast, A, is concerned about its market position relative to its major competitor, B, and to its other competitors, C. Their analysts believe that, in the short run, the major changes to the number of customers are due to brand switching. In particular, research indicates that, from month-to-month, 70% of A's customers stick with A, 25% switch to B, and the remainder switch to C. Of B's customers, 20% switch to A, 75% remain with B, and 5% switch to C. Of the customers in category C, 10% switch to A, 10% switch to B, and 80% remain with C. What happens to the evolution of this system over time?

i. We have the three coupled difference equations:
   1. $a_{n+1} = 0.7a_n + 0.2b_n + 0.1c_n$.
   2. $b_{n+1} = 0.25a_n + 0.75b_n + 0.1c_n$.
   3. $c_{n+1} = 0.05a_n + 0.05b_n + 0.8c_n$.

ii. Show evolution over time. Implement transition as a matrix.

iii. It appears that the final population ratios are independent of the initial population.

Homework:

- 1.1 #3abd, 13
- 1.2 #2
- 1.3 #4acd, 14 (use the model from Example 3 in Section 1.2)
- 1.4 #6
- Solve the following recurrences:
  - $a_{n+1} = a_n + 12a_{n-1}$, with $a_0 = 0$, $a_1 = 7$.
  - $a_{n+1} = 6a_n - 9a_{n-1}$, with $a_0 = 1$, $a_1 = 9$.
  - $a_{n+1} = (n+1)a_n + (n+1)!$, with $a_0 = 0$.